

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	Not Provided
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the main functional features implemented in Smart Lender. The project combines machine learning, web interaction, batch processing, session tracking, and reporting to support loan-screening use cases.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Single Applicant Loan Prediction	Accepts applicant financial and demographic inputs and predicts loan approval or rejection.	app.py , templates/index.html , templates/result.html	Done	High

2	JSON Prediction API	Exposes machine - learning inference through <code>/api/predict</code> for programmatic integration.	<code>app.py</code>	Done	High
3	EMI Calculator API	Calculates EMI, total payment, interest, and DTI using <code>/api/emi</code> .	<code>app.py</code>	Done	Medium
4	Batch CSV Upload and Scoring	Allows uploading CSV files to predict multiple applications at once.	<code>app.py</code> , <code>templates/batch.html</code>	Done	High
5	Batch Result Download	Generates downloadable CSV output for scored batch uploads.	<code>app.py</code>	Done	Medium
6	Session-Based Prediction History	Stores recent prediction records within the active user session and displays them on the history page.	<code>app.py</code> , <code>templates/history.html</code>	Done	Medium

7	Model Analytics and Reports	Displays model comparison information, feature importances, and generated reports/charts.	app.py , templates/about.html , templates/reports.html , static/reports	Done	Medium
8	Model Training and Selection Pipeline	Cleans data, engineers features, trains multiple classifiers, selects the best model, and stores artifacts for deployment.	train_model.py, model/	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Real-time loan prediction, model training pipeline, batch CSV scoring, JSON API access, and session history tracking

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home application form, result page, batch upload page, history/report views

2	Backend / Logic	3	Prediction workflow, preprocessing pipeline, EMI computation
3	Database / Storage	2	Model artifact storage in model/, temporary session storage for history and batch CSV output
4	API / Integration	3	/api/predict, /api/emi, /health
5	Security / Authentication	1	Basic Flask session secret and input validation; no dedicated login or authentication module is implemented